Glossary

3' Hydroxy end: The hydroxyl group attached to 3' carbon atom of sugar of the terminal nucleotide of a nucleic acid.

Bacterial artificial chromosomes (BAC): A cloning vector for isolation of genomic DNA constructed on the basis of F-factor.

Chimeric DNA: A recombinant DNA molecule containing unrelated genes.

Cleave: To break phosphodiester bonds of dsDNA, usually with a restriction enzyme.

Cloning site: A location on a cloning vector into which DNA can be inserted.

Cloning: Incorporation of a DNA molecule into a chromosomal site or a cloning vector.

Cloning Vector: A small, self-replicating DNA inserted in a cloning gene.

COS sites: The 12-base, single strand, complementary extension of phage lambda (l) DNA.

DNA Polymerase: An enzyme that catalyses the phosphodiester bond in the formation of DNA.

Endonucleases: An enzyme that catalyses the cleavage of DNA at internal position, cutting DNA at specific sites.

Genome: The entire complement of genetic material of an organism.

Insert DNA: A DNA molecule incorporated into a cloning vector.

Ligase: An enzyme used in genetic engineering experiment to join the cut ends of dsDNA.

M-13: AssDNA bacteriophage used as vector for DNA sequencing.

Phagemid: A cloning vector that contains components derived from both phage DNA and plasmid.

Plasmid: Extrachromosomal, self-replicating, circular dsDNA containing some non-essential genes.

Restriction map: A linear array of sites on DNA cleaved by various restriction enzymes.

Shuttle Vector: A plasmid cloning vector that can replicate in two different organisms due to the presence of two different origin of replication Ori^{EUK} and $Ori^{E.\ coli}$

Taq polymerase: A heat stable DNA polymerase isolated from a thermophilic bacterium *Thermus aquaticus.*

Vectors: Vehicles for transferring DNA from one cell to another.

Biofuel: Fuels like hydrogen, ethanol and methanol produced from a biological source by the action of microorganisms.

Bioleaching: Process of using microorganisms to recover metals from their ores or contaminant environment

Bioremediation: Process of using organisms to remove or reduce pollutants from the environment.

Green Technology: Pollution-free technology in which pollution is controlled at source.

Phytoremediation: Use of certain plants to remove contaminants or pollutants from the environment (soil, water or air).

Recombinant: Cell / Organism formed by a recombination of genes.

Transformation: Process of transferring a foreign DNA into a cell and changing its genome.

Vector: Agent used in recombinant DNA technique to carry new genes into foreign cells.

Wild Type: Natural form of organisms.

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